# Joey van Haren

Games Programming student at BUas · Gameplay & AI Systems

joeyvharen55@gmail.com · linkedin.com/in/joey-van-haren · chadjoey.github.io

## PROFILE

Third-year Games Programming student at Breda University of Applied Sciences, specializing in gameplay and AI systems. I started in web development, moved into game development through an MBO, and did an internship at ChimpWorks Games before enrolling at BUas. Since my second year I have focused on AI programming: building systems that feel responsive and intelligent, from stealth guards and cat AI to Monte Carlo decision making. My goal is to continue growing in that direction and eventually work in the AAA industry.

## PROJECTS

**Chasing Whiskers** · Unreal Engine 5, C++ 2026 – Present

Comedy action game in development for Steam. Responsible for the full cat AI system.

- Designed and implemented a six-state FSM covering roaming, caution, fleeing, hiding, following, and patrolling
- Built Lévy flight movement for naturalistic roaming and directionally biased fleeing
- Integrated Unreal nav links with custom ballistic arc jump physics by reading engine source directly
- Implemented BOIDS-based steering with a settle system for group following behaviour
- Built a hiding spot selection system with probability falloff by distance and movement direction
- Created a patrol path editor tool using Unreal's FComponentVisualizer with in-viewport waypoint dragging
- Exposed all parameters through data assets so designers can tune cat types without touching code

**Nakon** · Custom C++ Engine (Firewasp) 2025

First-person horde shooter inspired by Call of Duty Zombies, built on a custom engine over 8 weeks.

- Built the interactables system: wall buys, perk machines, and enemy power-up drops
- Implemented three enemy variants with per-wave health scaling and head hitbox detection
- Extended crowd navigation with surface clamping to prevent agents drifting off navmesh edges

**Mahjong AI** · Unreal Engine 5, C++ 2026

- Built a lookup-table shanten calculator covering standard hands, seven pairs, and thirteen orphans
- Implemented flat Monte Carlo with determinization for decision making under hidden information
- Parallelised simulation across all CPU cores using Unreal's ParallelFor with deterministic per-task seeds

**Godot Behaviour Tree Plugin** · Godot 4, GDExtension, C++ 2025

- Built a visual graph editor, runtime execution system, and full set of standard node types as a GDExtension
- Supported both C++ and GDScript for condition and action nodes
- Resolved an undocumented Godot 4.3 serialization bug affecting custom GDExtension resource arrays

**FPS Engine** · Custom C++ (Firewasp), Team of 6 2025

- Integrated Recast/Detour into a custom engine with no documentation, working largely from source
- Built a reusable agent behaviour API with vision frustum, detection meter, pathfinding, and combat
- Shipped a COD Zombies navigation stress test and a full PayDay 2-style stealth demo
- Engine was selected as custom tech for the following year-group project

## EXPERIENCE

**Internship – Game Developer** · ChimpWorks Games2021

First professional game development experience. Contributed to live projects and gained grounding in studio workflow and tooling.

## EDUCATION

**BSc Game Technologies** (expected 2026) · Breda University of Applied Sciences2023 – Present

**MBO 4 Game Developer** · Technova College2018 – 2021

## SKILLS

C++ · C# · Unreal Engine 5 · Unity · Godot · Gameplay AI · Finite State Machines · Behaviour Trees · Monte Carlo · Pathfinding · Recast / Detour · Steering Behaviours · BOIDS · Flow Fields · Perforce · Git